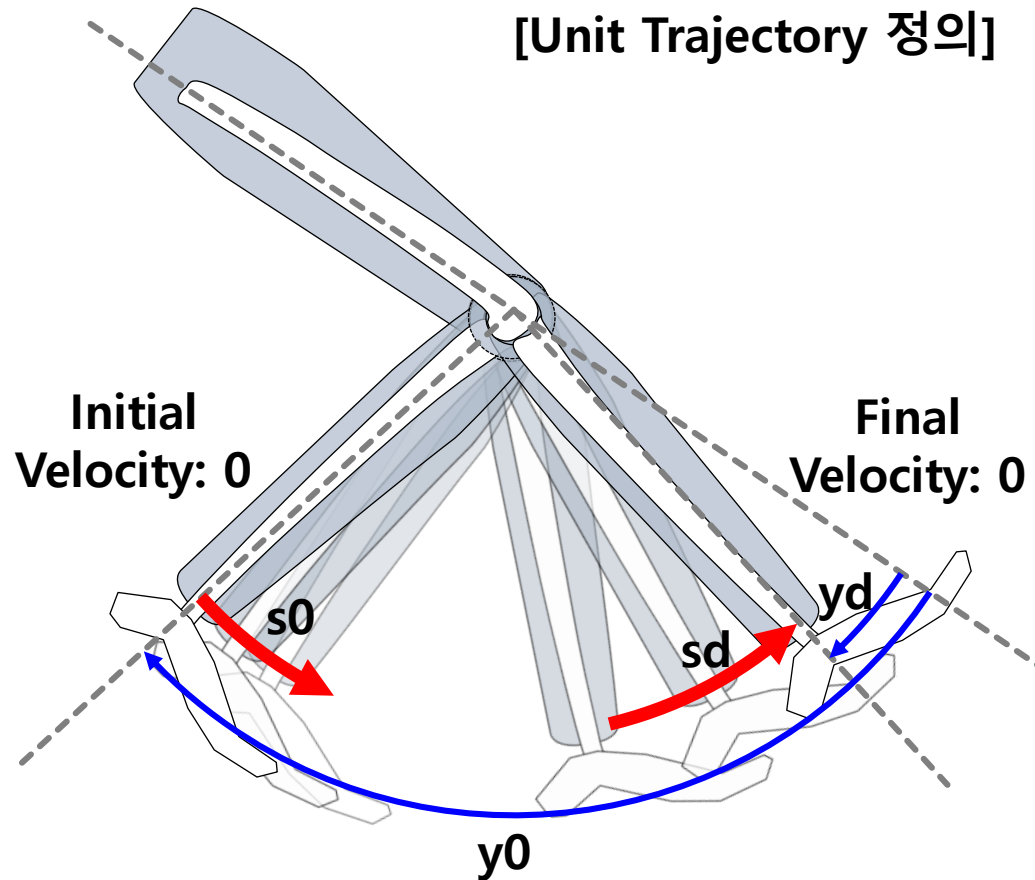


[Unit Trajectory 정의]



y_0 : 초기 위치

- Physical unit : degree (Output축 기준)
- Range : -360 ~ 360
- Neutral posture : 0
- Data format : int16

y_d : 목적 위치

- Unit and format : the same as y_0

s_0 : 가속 파라미터

- Physical unit : 무차원수
- Range : -128 ~ 128
- Data format : int8

s_d : 감속 파라미터

- Unit and format : the same as s_0

정지된 초기위치 y_0 에서, s_0 만큼 가속하면서 이동을 시작하고, 다시 s_d 만큼 감속하면서 목적 위치 y_d 까지 이동

P-Vector: Parameterized Position Trajectory

A unit trajectory, representing a unit movement, is expressed by a **5th-order high-order polynomial** as shown below

$$y(\tau) = a_0 + a_2\tau^2 + a_3\tau^3 + a_4\tau^4 + a_5\tau^5$$

Here, each variable and coefficient is defined as follows

$$\tau = \frac{k}{L_{trajectory}} \in [0,1], \quad k \in [0,1,2 \dots, L_{trajectory}]$$

$$a_0 = y_0$$

$$a_2 = 0.5s_0(y_d - y_0)$$

$$a_3 = (10 - 1.5s_0 + 0.5s_d)(y_d - y_0)$$

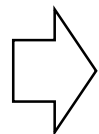
$$a_4 = (-15 + 1.5s_0 - s_d)(y_d - y_0)$$

$$a_5 = (6 - 0.5s_0 + 0.5s_d)(y_d - y_0)$$

즉, 주어진 y_0 에 대해 단위 동작은 $y_d, s_0, s_d, L_{trajectory}$ 의 함수
 $\rightarrow P(i) = [y_d, L_{traj}, s_0, s_d]$ 파라미터 벡터로 단위 궤적 정의 가능

- 촘촘한 픽셀 → 벡터로 변환
- SD카드 접근 →
[MotionMap.csv] File 확인

P(1) =	[200	1000	0	5];
P(2) =	[0	300	10	10];
P(3) =	[300	500	0	0];
P(4) =	[200	800	5	0];
P(5) =	[0	1500	10	10];
P(6) =	[400	2000	3	8];
P(7) =	[100	800	10	3];
P(8) =	[600	3000	20	20];
P(9) =	[200	1000	7	8];
P(10) =	[400	1000	12	6];

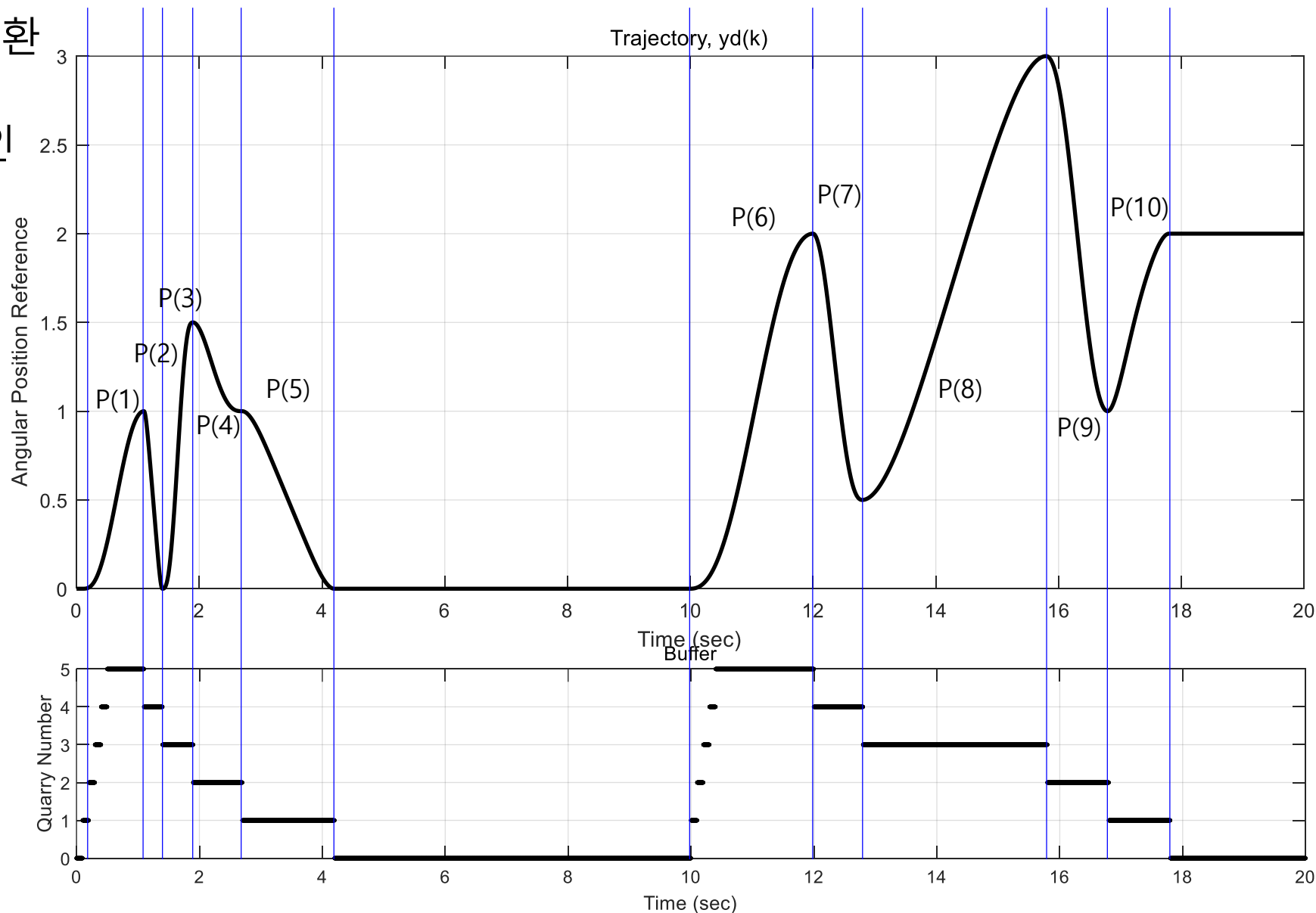


y_d

L_{traj} (궤적 길이)

s_0

s_d



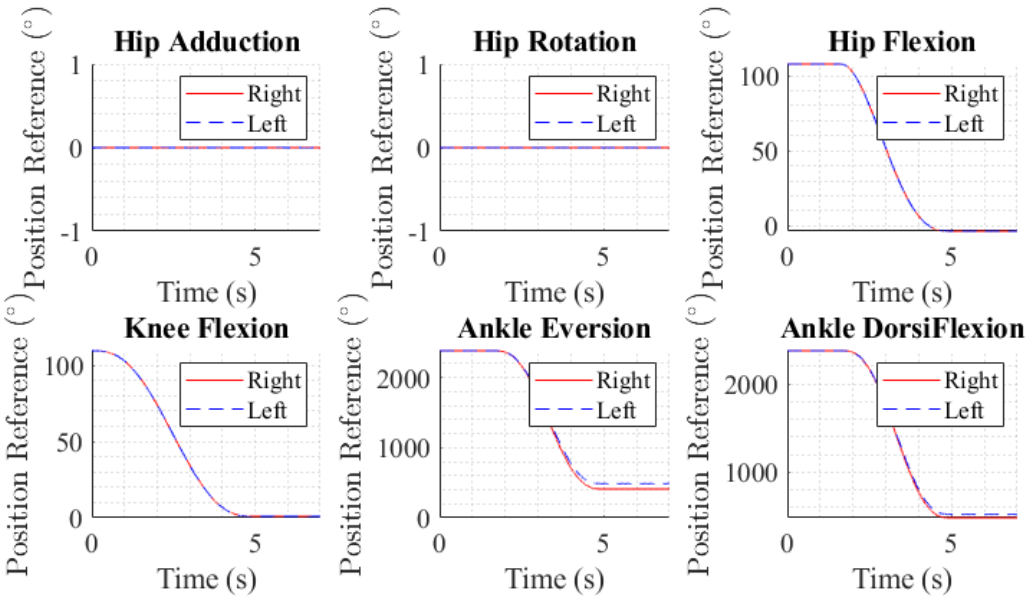
Motion Map (예시 : 일어서기)

MotionSet ID: *SIT2STAND*

“MotionMap.csv” file

MS ID	MS NAME	MD ID	C-Vector	P-Vector		
				1	2	
5	SIT2STAND	MD1	...	0, 5000, 0, 0	-	...
		MD2	...	0, 5000, 0, 0	-	...
		MD3	...	0, 5000, 0, 0	-	...
		MD4	...	0, 5000, 0, 0	-	...
		MD5	...	108, 1500, 0, 0	-3.6, 3333, 10, 0	...
		MD6	...	108, 1500, 0, 0	-3.6, 3333, 10, 0	...
		MD7	...	1, 5000, 0, 0	-	...
		MD8	...	1, 5000, 0, 0	-	...
		MD9	...	2389.4,1500.0,0.0,0.0	448.5446,3333.0,0.0,0.0	...
		MD10	...	2444.8,1500.0,0.0,0.0	531.2969,3333.0,0.0,0.0	...
		MD11	...	2396.3,1500.0,0.0,0.0	524.0605,3333.0,0.0,0.0	...
		MD12	...	2440.0,1500.0,0.0,0.0	567.2713,3333.0,0.0,0.0	...

Decoding
in MD



Motion Map (예시 : 걷기)

MotionSet ID: *NORMAL_WALKING*

"MotionMap.csv" file

MS ID	MS NAME	MD ID	C-Vector	P-Vector		
				1	2	
28	NORMAL_WALKING	MD1	...	322.2, 182.0, 13.8, 5.5	126.3, 390.0, 18.4, 1.2	...
		MD2	...	55.4, 184.0, 13.1, 5.9	207.2, 268.0, 8.0, 0.0	...
		MD3	...	0.5, 1020.0, 0.0, 0.0	-	...
		MD4	...	0.5, 1020.0, 2.5, 102.5	-	...
		MD5	...	45.7, 567.0, 3.7, 8.0	30.8, 453.0, 16.4, 0.0	...
		MD6	...	-2.7, 1020.0, 23.6, 28.8	-	...
		MD7	...	73.9, 488.0, 11.2, 7.3	23.4, 360.0, 4.3, 2.5	...
		MD8	...	9.0, 1020.0, 23.2, 27.5	-	...
		MD9	...	1375.2, 341.0, 0.5, 7.8	2072.3, 256.0, 9.7, 1.0	...
		MD10	...	1621.1, 387.0, 23.4, 3.5	1890.2, 308.0, 1.1, 4.1	...
		MD11	...	1392.8, 350.0, 1.3, 9.8	2034.5, 300.0, 15.3, 0.0	...
		MD12	...	2023.6, 447.0, 14.5, 0.3	2717.0, 573.0, 14.9, 33.1	...

Decoding
in MD

